Exact expected subgraph counts in the random k-out graph: closed forms, a matched-density Erdős–Rényi contrast, and an orientation invariant

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Abstract

In the *random k-out graph* $G(n, k\text{-out})$ each of the n vertices independently selects a k-subset of the other $n-1$ vertices, and two vertices are adjacent if either selected the other (symmetrized union). The model is a standard $O(nk)$-edge alternative to the Erdős–Rényi graph $G(n,p)$ and has been studied chiefly through global observables (connectivity, perfect matchings, Hamilton cycles). We develop its *local structure*. Our first result is an exact closed formula, valid for every fixed finite simple graph H and all finite (n,k), for the expected number of copies of H (Theorem 3.1); it relies on a per-chooser avoidance probability that is exact hypergeometric and on chooser independence, and it quantifies the *within-chooser negative correlation* $\Pr(v \in S_u,\, w \in S_u) = \frac{k(k-1)}{(n-1)(n-2)}$ that invalidates naive $G(n,p)$-style shortcuts. We then compute: (i) the exact triangle closed form $P(\triangle) = 1 - 3A^2 + 3A^2B - B^3$ and its series, the limiting mean $\mathbb{E}T \to (4k^3 - 3k^2)/3$, the limiting variance $\mathrm{Var}(T) \to (4k^3 - 3k^2)/3$ (with exact $1/n$ terms), and the exact $k = 1$ specials $P(\triangle) = 2/(n-1)^3$ (Theorems 4.1, 8.1); (ii) a matched-density comparison with $G(n,p)$: the expected triangle counts obey a ratio tending to $1 - \frac{3}{4k}$, with the four-cycle ratio tending to $1 - \frac{1}{k} + \frac{1}{8k^2}$ (Proposition 5.1); (iii) an exact cancellation theorem for stars $K_{1,m}$: the first non-zero term of the fixed-vertices star probability is exactly of order n^{-m}, and $\mathbb{E}[\,\#\text{ unlabelled } K_{1,m}\,] = \frac{c_m(k)}{m!}\, n\,(1 + o(1))$ with explicit $c_m(k)$ (Theorem 6.1); and (iv) a general asymptotic description: for any fixed H, $\mathbb{E}[\,\#\text{ labelled copies of } H\,] \sim \Omega_k(H)\, n^{v-e}$, where

$$\Omega_k(H) \;=\; \sum_{\sigma} \prod_{x \in V(H)} (k)_{t_x(\sigma)}$$

is an explicit *orientation invariant* summed over the assignments σ of each edge of H to exactly one of its endpoints, $(k)_t$ being a falling factorial (Theorem 7.1). $\Omega_k(H) = 0$ if and only if no assignment keeps every vertex's owned degree at most k. All approximate statements above carry machine-exact series constants. All closed forms are verified at exact precision: symbolically (sympy), by exhaustive enumeration over *all* choice profiles for small n (difference zero), and by Monte Carlo at large n (all $|z| \le 1.43$). The Poisson-type limit of the triangle count is stated as a conjecture supported by two exact moments; we do not claim it as a theorem.

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1 Introduction

1.1 The model and motivation

The *random k-out graph* $G(n, k\text{-out})$ (in the symmetrized form used throughout the literature, e.g. [1, Ch. 22], [10]) is constructed as follows. Each of the n vertices v independently and uniformly selects a k-subset $S_v \subseteq [n] \setminus \{v\}$; vertices u and v are adjacent if either $u \in S_v$ or $v \in S_u$. The resulting simple graph has exactly between $nk/2$ and nk edges ($\mathbb{E}[\#\text{edges}] = nk - \frac{nk^2}{2(n-1)}$ exactly, Proposition 2.2) — a degree-bounded, connected-with-high-probability ($k \geq 2$) sparse model.

$G(n, k\text{-out})$ is a workhorse model in situations where each node has a hard budget of $\approx k$ directed links: random pairwise key predistribution in sensor networks [13, 7, 10], payment channel networks, and related designs; see [9] and the references in [10]. Its *global* behaviour is well understood: connectivity is achieved almost surely for every $k \geq 2$ [2, 1], with sharp finite-n bounds [10]; perfect matchings and Hamilton cycles occur with high probability [1, Ch. 22]; and k-out subgraphs of large hosts are k-connected w.h.p. [3].

1.2 The gap: local structure

By contrast, the *local* structure — the expected number of copies of a fixed small graph H (triangle, 4-cycle, star, path, ...) at finite (n, k) — has not, to our knowledge, been developed. The chapter of [1] devoted to k-out graphs (Ch. 22) contains connectivity, perfect matchings, Hamilton cycles, and nearest-neighbour graphs, and no counts of fixed subgraphs (checked against the authors' full text; Section 10). The recent triangle analyses for other correlated models — random key graphs [7, 8], clustered random graphs [11], and the clustering-index estimates of [6] (which treat Erdős–Rényi expressions and simulate regular/Barabási–Albert/Watts–Strogatz models, without the k-out model) — do not cover this model either.

The question is non-trivial and the answer is not a transposition of the $G(n, p)$ computation, because of a genuine structural feature: the maker choices *within* one chooser are negatively correlated,

$$\Pr\big(v \in S_u \wedge w \in S_u\big) = \frac{k(k-1)}{(n-1)(n-2)} \;<\; \Big(\frac{k}{n-1}\Big)^2, \tag{1}$$

so the per-vertex choices do not behave as independent coin flips, and a naive $G(n, p)$-style "independence" computation can overshoot the truth by factors of several (Table 1: at $k = 1$, $n = 200$ the naive estimate is $3.97\times$ the exact value).

1.3 Contributions and overview

We establish, for every fixed finite simple graph H (v vertices, e edges) and all finite (n, k) with $1 \leq k \leq n - 1$:

1. **Exact general formula** (Theorem 3.1): an alternating per-edge inclusion–exclusion whose summands are exact products of per-chooser avoidance probabilities. It computes $\mathbb{E}[\#\{\text{labelled copies of } H\}]$ exactly at any (n, k).

2. **Triangles** (Theorems 4.1 and 8.1): the exact closed form $P(\triangle) = 1 - 3A^2 + 3A^2B - B^3$ (with $A = \frac{\binom{n-2}{k}}{\binom{n-1}{k}}$, $B = \frac{\binom{n-3}{k}}{\binom{n-1}{k}}$), an alternate identity exposing the negative correction to the naive value, exact series coefficients $(4k^3 - 3k^2)/3$, $-\frac{k^2(k-1)(4k-3)}{2}$, $\frac{k^2(6k^3-39k^2+46k-15)}{6}$, exact $k = 1$ specials ($P(\triangle) = 2/(n-1)^3$), and the exact second moment of the triangle count with $\mathrm{Var}(T) \to (4k^3 - 3k^2)/3$ and $1/n$-term $-\frac{k^2(16k^2-21k+7)}{2}$.

3. **Matched-density Erdős–Rényi contrast** (Proposition 5.1): with the edge density $q = 2p - p^2$, $p = \frac{k}{n-1}$, matched between the two models, the ratio of expected triangle counts tends to $1 - \frac{3}{4k}$ (e.g. $\frac{1}{4}$ at $k = 1$, $\frac{7}{8}$ at $k = 4$), with explicit $1/n$ corrections; the 4-cycle ratio tends to $1 - \frac{1}{k} + \frac{1}{8k^2}$.

4. **Stars** (Theorem 6.1): an exact closed form for stars $K_{1,m}$ and the exact orders: the first non-zero term of the star probability at fixed vertices is exactly of orders n^{-m}, so $\mathbb{E}[\#\text{unl. } K_{1,m}] = \frac{c_m(k)}{m!}\, n(1 + o(1))$ with explicit polynomials $c_m(k)$ up to $m = 6$ (Appendix, Table 3) and, in general, $c_m(k) = \sum_j \binom{m}{j} (k)_j\, k^{m-j}$.

5. **General asymptotics via an orientation invariant** (Theorem 7.1): for any fixed H,

$$\mathbb{E}[\#\{\text{labelled copies of } H\}] \,/\, n^{v-e} \;\longrightarrow\; \Omega_k(H) \;=\; \sum_{\sigma} \prod_{x} (k)_{t_x(\sigma)},$$

   the sum over the 2^e assignments σ of each edge to exactly one of its endpoints, with $t_x(\sigma)$ the number of edges assigned to x. $\Omega_k(H) = 0$ iff no assignment gives every vertex owned degree at most k; the proof is a two-sided Bonferroni argument and is complete.

6. **Verification stack** (Section 9): all closed forms cross-checked symbolically over the rationals, by exhaustive enumeration over *all* choice profiles at small n (agreement to machine zero, e.g. $\mathrm{Var}(T) = \frac{15985}{5832}$ at $(n, k) = (5, 2)$), and by Monte Carlo at large n (Table 4: worst $|z| = 1.43$ on means, 1.41 on variances).

Honesty statement on novelty. We are *not* aware of any published closed-form expression for the expected number of copies of a fixed graph H (with $e \geq 2$) in the symmetrized random k-out graph, at finite (n, k) or in asymptotic form. This negative statement is the result of the targeted search described in Section 10; we cannot exclude the existence of unindexed or overlooked prior work, and we phrase our contribution accordingly.

1.4 Notation

Throughout, $n \geq 2$, $1 \leq k \leq n - 1$, and $(a)_m = a(a - 1) \cdots (a - m + 1)$ is the falling factorial (with $(a)_0 = 1$), $\binom{r}{s} = 0$ if $r < s$ or $s < 0$. All labels are of the template H's vertices; a *labelled copy* of H in G is an injective map $\varphi : V(H) \to [n]$ such that $\varphi(u)\varphi(v) \in E(G)$ for every $uv \in E(H)$; the number of *unlabelled* copies sitting on a given vertex set is the labelled count divided by $|\mathrm{Aut}(H)|$. The expectation is over the random chooser sets $(S_v)_{v \in [n]}$, independent across vertices.

2 Preliminaries

Definition (Model)**.** $G(n, k\text{-out})$: every $v \in [n]$ draws S_v uniformly from the $\binom{n-1}{k}$ k-subsets of $[n] \setminus \{v\}$; the chooser sets (S_v) are independent. An edge uv exists iff $u \in S_v$ or $v \in S_u$.

For a fixed unordered pair $\{u, v\}$, $\Pr(uv \in E) = q := 2p - p^2$ with $p = \frac{k}{n-1}$: either v chose u (probability p) or else v did not choose u (prob. $1 - p$) and u chose v (prob. p, independent across choosers). Note $q = 1 - A^2$ with $A := \binom{n-2}{k} / \binom{n-1}{k} = (1 - \frac{k}{n-1})$.

Lemma 2.1 (Within-chooser negative correlation)**.** *For fixed u and distinct $v, w \neq u$,*

$$\Pr(v \in S_u \wedge w \in S_u) = \frac{\binom{n-3}{k-2}}{\binom{n-1}{k}} = \frac{k(k-1)}{(n-1)(n-2)} < \left(\frac{k}{n-1}\right)^2 = \Pr(v \in S_u) \Pr(w \in S_u).$$

Proof. Hypergeometric: the count of k-subsets of $[n] \setminus \{u\}$ containing both v, w is $\binom{n-3}{k-2}$. The ratio expands to $\frac{k(k-1)}{(n-1)(n-2)}$, and $\frac{k(k-1)}{(n-1)(n-2)} < \frac{k^2}{(n-1)^2}$ for $k \geq 1$. □ □

Chooser independence (across different vertices u) is exact; it is only the *within*-chooser dependencies (Lemma 2.1) that break the $G(n,p)$ analogy. Our exact formulae keep track of them through exact avoidance probabilities.

Proposition 2.2 (Edges)**.**

$$\mathbb{E}[\,\#\textit{edges}\,] = \binom{n}{2} q = nk - \frac{nk^2}{2(n-1)} \quad \textit{exactly.}$$

3 The general subgraph-count formula

Fix a template H with vertex set $\{1, \ldots, v\}$ and edge list $E(H)$. For a subset $F \subseteq E(H)$ let $d_x(F)$ be the degree of x in the spanning subgraph $(V(H), F)$.

Theorem 3.1 (Exact expected labelled copy count)**.** *For fixed (n,k) and any fixed injective map $\varphi : V(H) \to [n]$,*

$$\Pr\big(\varphi \textit{ is an embedding}\big) \;=\; \sum_{F \subseteq E(H)} (-1)^{|F|} \prod_{x \in V(H)} \frac{\binom{n-1-d_x(F)}{k}}{\binom{n-1}{k}},$$

hence, exactly,

$$\mathbb{E}\big[\,\#\{\textit{labelled copies of } H\}\,\big] = (n)_v \sum_{F \subseteq E(H)} (-1)^{|F|} \prod_{x} \frac{\binom{n-1-d_x(F)}{k}}{\binom{n-1}{k}},$$

where the convention $\binom{r}{k} = 0$ for $r < k$ applies. For fixed k and H, the summands are rational numbers (exact), and the formula involves at most 2^e terms.

Proof. By symmetry over injective maps, the probability for any fixed φ equals the probability for the identity embedding; write x for the vertex of label $\varphi(x)$. For $uv \in E(H)$ let X_{uv} be the event that uv is an edge of $G(n,k\text{-out})$. Then $\Pr(\forall uv \in E(H) : X_{uv}) = \sum_{F \subseteq E(H)} (-1)^{|F|} \Pr\big(\bigcap_{uv \in F} \overline{X_{uv}}\big)$ by inclusion–exclusion.

Now $\bigcap_{uv \in F} \overline{X_{uv}}$ holds iff every chooser x avoids each of its $d_x(F)$ template-neighbours. The chooser sets are independent across x, and the probability that x's uniformly random k-subset of $[n] \setminus \{x\}$ avoids a fixed set of $d_x(F)$ forbidden targets is exactly $\binom{n-1-d_x(F)}{k} / \binom{n-1}{k}$ (hypergeometric, exact); multiplying this over the (independent) choosers $x \in V(H)$ gives $\prod_x \binom{n-1-d_x(F)}{k} / \binom{n-1}{k}^v$. The expectation follows because for a uniform injective φ, the φ-images of the v template vertices are the entries of a uniformly random v-tuple of distinct vertices, giving the factor $(n)_v$. □ □

Remark. The formula has the shape of a $G(n,p)$-type inclusion–exclusion *except* that the "per-edge exponentially-closed" factors of $G(n,p)$ (factors $(1-p^2)^{|\cdot|}$) are replaced by *per-chooser* products of exact avoidance probabilities; the difference is precisely what survives of the negative correlation of Lemma 2.1. For later use we also record that all predictions of Theorem 3.1 were verified independent of it, by exhaustive enumeration over all choice profiles (Section 9: every comparison has absolute difference $\leq 2.2 \cdot 10^{-15}$, mostly exactly 0).

4 Triangles: exact formula, asymptotics, and the naive gap

Let T be the number of (unlabelled) triangles: $T = \sum_{\{a,b,c\}} \mathbf{1}\{abc \text{ triangle}\}$.

Theorem 4.1 (Exact triangle probability)**.** *Let*

$$A = \frac{\binom{n-2}{k}}{\binom{n-1}{k}} = 1 - \frac{k}{n-1}, \qquad B = \frac{\binom{n-3}{k}}{\binom{n-1}{k}} = \Big(1 - \frac{k}{n-1}\Big)\Big(1 - \frac{k}{n-2}\Big).$$

Write $C := \frac{n-2-k}{n-2}$*, so that* $B = AC$*. Then exactly*

$$P(\triangle) := \Pr(\textit{fixed triple } \{a,b,c\} \textit{ spans a triangle}) = 1 - 3A^2 + 3A^2B - B^3, \tag{2}$$

and, equivalently,

$$P(\triangle) \;=\; \underbrace{(1-A^2)^3}_{\textit{naive } G(n,p) \textit{ value}} \;+ A^3\,(C-A)\,\big(3 - A^2 - AC - C^2\big), \tag{3}$$

where the second term is ≤ 0 *(since* $C < A$*), a* negative correction *to the naive value.*

Proof. By Theorem 3.1 specialized to the triangle (edges ab, bc, ca), a subset $F \subseteq \{ab, bc, ca\}$ of size $0, 1, 2, 3$ yields (d_a, d_b, d_c) equal to $(0,0,0)$; $(1,1,0)$-permutations; $(2,1,1)$-permutations; and $(2,2,2)$, respectively. There are 1, 3, 3, 1 such subsets, hence $P(\triangle) = 1 - 3A^2 + 3A^2B - B^3$, since the middle product is $\frac{\binom{n-3}{k}}{\binom{n-1}{k}} \cdot \frac{\binom{n-2}{k}}{\binom{n-1}{k}} \cdot \frac{\binom{n-2}{k}}{\binom{n-1}{k}} = B\,A^2$. The identity (3) is an algebraic rearrangement using $B = AC$ and $q = 1 - A^2$; both the relation $B \equiv AC$ and the equivalence (3)≡(2) are machine-checked symbolically. The sign of the correction: $C - A = \frac{n-2-k}{n-2} - \frac{n-1-k}{n-1} = -\frac{k}{(n-1)(n-2)} < 0$, while $3 - A^2 - AC - C^2 > 0$ for the relevant range (it is $3 - (A^2 + AC + C^2)$ with $A, C \in (0,1)$, so $A^2 + AC + C^2 < 3$). □ □

Theorem 4.2 (Asymptotics of the mean)**.** *As* $n \to \infty$ *with* k *fixed, with* T *the triangle count,*

$$\mathbb{E}T \;=\; e_0 + \frac{e_1}{n} + \frac{e_2}{n^2} + O\!\left(\frac{1}{n^3}\right),$$

where

$$e_0 = \frac{4k^3 - 3k^2}{3}, \qquad e_1 = -\frac{k^2(k-1)(4k-3)}{2}, \qquad e_2 = \frac{k^2(6k^3 - 39k^2 + 46k - 15)}{6},$$

and the next coefficient is $e_3 = -\frac{k^2(k^4 - 27k^3 + 99k^2 - 96k + 27)}{6}$ *(by which the truncated series at* $1/n^2$ *is controlled). In particular* $\lim \mathbb{E}T = (4k^3 - 3k^2)/3$*, i.e.* $\Theta(1)$ *for fixed* k*: the mean triangle count does not grow with* n *at all.*

The coefficients were derived by exact series expansion of $\binom{n}{3}P(\triangle)$ (Theorem 4.1) in $1/n$ and are machine-checked symbolically; the truncation at e_2 was additionally checked against exact Fraction evaluations at $n = 400$ for $k = 1, \ldots, 5$, with residuals bounded by $|e_3|/n^3$ (Section 9).

Corollary 4.3 (Exact $k = 1$ specials)**.** *For* $k = 1$*:*

$$P(\triangle) = \frac{2}{(n-1)^3} \quad \textit{and} \quad P(\textit{4-cycle}) = \frac{2}{(n-1)^4} \quad \textit{for all } n.$$

These were simplified to closed form from the general formulae and machine-checked symbolically.

Table 1: Exact $\mathbb{E}T$ vs. the naive ($G(n,p)$-style) prediction $(1-A^2)^3\binom{n}{3}$ at $n = 200$. From `results/expectations_table.csv`.

k	exact $\mathbb{E}T$	naive	rel. error	naive/exact
1	0.333325	1.323275	2.9699	3.97
2	6.616147	10.506404	0.5880	1.59
3	26.592529	35.191162	0.3233	1.32
4	67.768683	82.784150	0.2216	1.22
5	137.416009	160.459789	0.1677	1.17
6	242.574103	275.163298	0.1343	1.13

Remark (Exact form of the naive gap)**.** The naive $G(n,p)$-style prediction for a triple is $q^3 = (1-A^2)^3$; the exact gap is, by identity (3),

$$q^3 - P(\triangle) = -A^3(C-A)\big(3 - A^2 - AC - C^2\big) \;\geq\; 0,$$

a rational function of (n,k) that starts contributing only from order n^{-5} (the orders n^{-3}, n^{-4} coefficients of the gap are 0); its first non-zero terms are $\frac{6k^2}{n^5} - \frac{3k^2(7k-9)}{n^6} + \cdots$. Nevertheless the *relative* error of the naive prediction is large because the truth itself is small; see Table 1.

At matched edge density q (same expected edges as $G(n,p)$), the exact $1/n$-expansion of the ratio is

$$\frac{P(\triangle)}{q^3} = \underbrace{1 - \frac{3}{4k}}_{r_0} + \underbrace{\frac{3(4k-3)}{8k}}_{r_1}\frac{1}{n} + O(n^{-2}),$$

which is analysed next (Proposition 5.1).

5 The Erdős–Rényi contrast at matched density

The natural comparison uses the same expected edge density $q = 2p - p^2$, $p = \frac{k}{n-1}$, in the Erdős–Rényi model $G(n,q)$, for which $\mathbb{E}T_{ER} = \binom{n}{3}q^3$ exactly.

Proposition 5.1 (Triangle and 4-cycle ratios)**.** *As $n \to \infty$ with $k \geq 1$ fixed:*

$$\rho_\triangle := \frac{\mathbb{E}T_{k\text{-}out}}{\mathbb{E}T_{ER}} = \frac{P(\triangle)}{q^3} \;\longrightarrow\; 1 - \frac{3}{4k},$$

with $1/n$-term $r_1 = \frac{3(4k-3)}{8k}$ and next term $-\frac{3(3k^2-21k+14)}{16k}\cdot n^{-2}$; and

$$\rho_{C_4} := \frac{\mathbb{E}[\#C_4]_{k\text{-}out}}{\mathbb{E}[\#C_4]_{ER}} \;\longrightarrow\; 1 - \frac{1}{k} + \frac{1}{8k^2},$$

with $1/n$-term $\frac{8k^2-8k+1}{4k^2}$.

Proof. $P(\triangle)/q^3$ and the closed form for $P(C_4)$ (computed in Lemma 5.2 below) divided by q^4 are rational functions of (n,k). The expansions are exact series expansions of those two ratios in $1/n$ (machine-checked symbolically; the leading terms simplify to the stated polynomial expressions in k). □ □

Lemma 5.2 (4-cycle closed form)**.**

$$P(C_4) = 1 - 4A^2 + 4A^2B + 2A^4 - 4A^2B^2 + B^4,$$

for a fixed labelled 4-cycle $\{abcd\}$, with $\mathbb{E}[\#\text{labelled } C_4] = (n)_4 P(C_4)$ and $\mathbb{E}[\#\text{unlabelled } C_4] = 3\binom{n}{4}P(C_4)$.

Table 2: Triangles: exact series vs. limits ($n = 1600$ row of `results/asymptotic_convergence.csv`).

k	$\mathbb{E}_{n=1600}$	e_0	$(\mathbb{E} - e_0)\, n$	e_1	$\rho_\triangle$ ($\to 1 - \frac{3}{4k}$)
1	0.3333332	$\frac{1}{3}$	$-2.1 \cdot 10^{-4}$	0	0.250235 (0.250)
2	6.6604086	$\frac{20}{3}$	-10.0129	-10.0000	0.625587 (0.625)
3	26.9493363	27	-81.0619	-81.0000	0.750704 (0.750)
4	69.1382595	$\frac{208}{3}$	-312.1181	-312.0000	0.813262 (0.8125)
5	141.1354011	$\frac{425}{3}$	-850.0249	-850.0000	0.850797 (0.850)

Proof. Apply Theorem 3.1 to the template C_4; the signed sum over the $2^4 = 16$ subsets groups into classes by edge pattern (verified symbolically: the Theorem 3.1 sum equals the closed form as an identity in (n, k)). □ □

5.1 Interpretation

For every fixed $k \geq 1$, the k-out graph has *asymptotically* $\left(1 - \frac{3}{4k}\right)$-fraction of the triangles that an Erdős–Rényi graph of equal expected edge density has: $\frac{1}{4}$ at $k = 1$, $\approx \frac{2}{3}$ approaching as $k \to \infty$. This is a large, real effect in the opposite direction from most "good clustering" enhancers (Barabási–Albert, Watts–Strogatz): the k-out budget spreads links thin and thereby makes the neighbourhoods of vertices locally tree-like, and the effect is *stronger for small* k. Together with the exact finite-n formulae, the constants $1 - \frac{3}{4k}$ and $1 - \frac{1}{k} + \frac{1}{8k^2}$ quantify what practitioners trade away (or gain, for triangle-suppression purposes) when replacing an Erdős–Rényi null by a degree-bounded k-out null.

6 Stars: exact cancellation and explicit constants

We use $K_{1,m}$ (a star with m leaves) at *fixed* vertices $\{u$ (center), $v_1, \ldots, v_m$ (leaves)$\}$ and $N_{K_{1,m}}$ for the count of unlabelled copies.

Theorem 6.1 (Stars)**.** *Fix $m \geq 2$ and $k \geq 1$. Exactly,*

$$P(K_{1,m}) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{\binom{n-1-j}{k}}{\binom{n-1}{k}} \left(\frac{\binom{n-2}{k}}{\binom{n-1}{k}}\right)^j, \tag{4}$$

and the first non-zero term of $P(K_{1,m})$ in powers of $1/n$ is exactly of order n^{-m}*, with finite non-zero constant*

$$\lim_{n\to\infty} n^m P(K_{1,m}) \;=\; c_m(k) \;:=\; \sum_{j=0}^{m} \binom{m}{j} (k)_j \, k^{m-j} \;>\; 0. \tag{5}$$

Consequently

$$\mathbb{E}\big[\, N_{K_{1,m}} \big] = \frac{(n)_{m+1}}{m!} P(K_{1,m}) = \frac{c_m(k)}{m!}\, n \left(1 + o(1)\right) \quad \text{as } n \to \infty,$$

with explicit polynomial constants $c_2(k) = k(4k-1)$, $c_3(k) = 2k(4k^2 - 3k + 1)$, $c_4(k) = k(16k^3 - 24k^2 + 19k - 6)$, $c_5(k) = 2k(16k^4 - 40k^3 + 55k^2 - 40k + 12)$, $c_6(k) = k(64k^5 - 240k^4 + 500k^3 - 615k^2 + 418k - 120)$ (Table 3).

Table 3: Stars: $n^m P(K_{1,m})$ converging to $c_m(k)$ ($n = 1600$ rows of `results/star_cancellation.csv`; residuals $O(1/n)$).

m	k	$n^m P$ at $n = 1600$	$c_m(k)$	$\mathbb{E}[N_{K_{1,m}}]$ at $n = 1600$
2	2	14.0037	14	11182.0
2	3	32.9887	33	26341.5
3	2	44.0300	44	11697.3
3	3	167.9715	168	44624.6
4	2	128.1500	128	8490.1
4	3	801.1104	801	53074.3
5	2	352.6002	352	4657.4
5	3	3619.9307	3618	47814.9
6	2	930.1018	928	2039.9
6	3	15648.4509	15633	34320.3

Proof. (4) is the specialization of Theorem 3.1 to the star: a subset F of the m star edges consists of j edges, all incident to the center; the spanning degrees are $d_{\text{center}} = j$ and $d_{v_i} = 1$ for the j selected leaves and 0 for the others. Hence each F of size j contributes sign $(-1)^j$ times the product $\frac{\binom{n-1-j}{k}}{\binom{n-1}{k}} \left(\frac{\binom{n-2}{k}}{\binom{n-1}{k}}\right)^j$, and there are $\binom{m}{j}$ such subsets, summing to (4). The identity between (4) and the general formula is machine-checked for $m = 2, \ldots, 6$ (as identities in (n, k) over the rationals).

For (5): the orientation-invariant Theorem 7.1 below (stated and proved in Section 7) applies with $H = K_{1,m}$, $v = m+1$, $e = m$, so $v - e = 1$: $\mathbb{E}[\#\text{labelled copies}] = \Omega_k(K_{1,m})\, n\, (1 + o(1))$ with $\Omega_k(K_{1,m}) = \sum_j \binom{m}{j} (k)_j k^{m-j}$ (each edge is assigned to either its center or its leaf; assignments with j center-owned edges contribute $\binom{m}{j}(k)_j k^{m-j}$). Dividing by $|\mathrm{Aut}(K_{1,m})| = m!$ and using $\mathbb{E}[N_{K_{1,m}}] = (n)_{m+1} P(K_{1,m})/m!$ yields $n^m P(K_{1,m}) \to c_m(k)$, which is finite and positive: all summands of (5) are non-negative and the $j = 0$ summand is $k^m > 0$. □

Remark. We emphasize two points of honesty about Theorem 6.1. (a) The identity (5), with the closed-form constants verified numerator-by-numerator in `src/experiments.py` (`STAR_C`, $m \le 6$), rests on Theorem 7.1, whose proof is complete in this paper (Section 7). (b) The exact orders claim — that *no* term of order between n^0 and n^{-m} survives in (4) — *follows* from (5) with $c_m(k) > 0$ finite: any lower-order non-zero term would blow up $n^m P(K_{1,m})$. So the exact-order statement is *equivalent* to the finite-non-zero-limit statement, and both carry the weight of Theorem 7.1. The $m \le 6$ cases were in addition directly series-expanded symbolically, and the constants $c_m(k)$ assessed numerically with the residual decaying like $1/n$ (Table 3; e.g. $n^6 P(K_{1,6})$ at $k = 2$: 930.1018 at $n = 1600$ vs. $c_6(2) = 928$).

7 The orientation invariant: general asymptotics

Fix a template H with v vertices and e edges. An *edge-owner assignment* is a function $\sigma : E(H) \to V(H)$ with $\sigma(uv) \in \{u, v\}$ for every edge uv; let $t_x(\sigma) = |\sigma^{-1}(x)|$ be the number of edges assigned to x. Assignments correspond bijectively to *orientations* of H's edges (towards the fixed endpoint "owner", or equivalently edge → one of its two endpoints). Write $(k)_t$ for the falling factorial.

Theorem 7.1 (Orientation invariant)**.** *For every fixed finite simple H and fixed $k \ge 1$,*

$$\mathbb{E}\big[\,\#\{\textit{labelled copies of } H\}\,\big] = \Omega_k(H)\, n^{v-e}\,(1 + o(1)) \quad \textit{as } n \to \infty,$$

where

$$\Omega_k(H) \;=\; \sum_{\sigma\in\{0,1\}^{E(H)}} \;\prod_{x\in V(H)} (k)_{t_x(\sigma)}.$$

Moreover, $\Omega_k(H)=0$ if and only if no *assignment σ satisfies $t_x(\sigma)\le k$ for all x; equivalently, iff every orientation of H leaves some vertex owning $\ge k+1$ edges; equivalently (via the orientation identity) iff H cannot be oriented such that each vertex has out-degree at most k — i.e. H admits no orientation of maximum owned-degree $\le k$.*

Proof. Fix the template labels (the identity embedding, by symmetry). For an assignment σ let Y_σ be the event that, for every edge $uv\in E(H)$, the *owner* $\sigma(uv)$ has the *other* endpoint of uv in its chosen set: if uv is owned by u, this means $v\in S_u$. If Y_σ holds, then every template edge is present (each edge is chosen by its owner), so $Y_\sigma\subseteq\{\text{all template edges present}\}$; conversely, if all edges are present, then every edge has at least one choosing endpoint, and choosing one such endpoint per edge extends to an assignment; hence $\{\text{all template edges present}\}=\bigcup_\sigma Y_\sigma$.

The chooser sets are independent across vertices, and the probability that a particular chooser x places at least its $t_x(\sigma)$ prescribed target-neighbours into its k-subset is exactly $(k)_{t_x(\sigma)}/(n-1)_{t_x(\sigma)}$ (the count of k-subsets containing t specific targets divided by $\binom{n-1}{k}$). Hence, by independence,

$$\Pr(Y_\sigma) \;=\; \prod_x \frac{(k)_{t_x(\sigma)}}{(n-1)_{t_x(\sigma)}}\,.$$

Since $\sum_x t_x(\sigma)=e$, we get

$$(n)_v\,\Pr(Y_\sigma)=\frac{(n)_v}{\prod_x(n-1)_{t_x(\sigma)}}\;\prod_x(k)_{t_x(\sigma)}=n^{v-e}\,\Omega(\sigma)\,\big(1+O(n^{-1})\big),$$

uniformly in σ, where $\Omega(\sigma)=\prod_x(k)_{t_x(\sigma)}$. Summing over the 2^e assignments:

$$(n)_v\,\Pr\Big(\bigcup_\sigma Y_\sigma\Big)=\Omega_k(H)\,n^{v-e}\,\big(1+O(n^{-1})\big)\;\;+\;\;\text{(over-count correction)}.$$

Two-sided control of the correction: for $\sigma\neq\tau$, the event $Y_\sigma\cap Y_\tau$ requires at least one edge uv to be chosen by both endpoints (the edge on which they disagree must lie in both), an event of probability $\frac{k(k-1)}{(n-1)(n-2)}=O(n^{-2})$; the remaining $e-1$ or fewer edges contribute witnesses at $\ge n^{-1}$ each, so $\Pr(Y_\sigma\cap Y_\tau)=O(n^{-e-1})$. There are at most $\binom{2^e}{2}=O(1)$ pairs (σ,τ), hence by the second Bonferroni inequality (applied to $\bigcup_\sigma Y_\sigma$),

$$\Big|\,\Pr\Big(\bigcup_\sigma Y_\sigma\Big)-\sum_\sigma\Pr(Y_\sigma)\Big|\;\le\;\sum_{\sigma<\tau}\Pr\big(Y_\sigma\cap Y_\tau\big)=O(n^{-e-1}),$$

so $(n)_v\Pr(\cup Y_\sigma)=\Omega_k(H)\,n^{v-e}(1+o(1))+O(n^{v-e-1})$, which is the claim.

The "iff" statement: $\Omega_k(H)=\sum_\sigma\prod_x(k)_{t_x(\sigma)}$ is a sum of non-negative terms. If some assignment σ satisfies $t_x(\sigma)\le k$ for every x, then all its factors $(k)_{t_x(\sigma)}$ are positive integers (each at least 1), so its summand is ≥ 1 and $\Omega_k(H)\ge 1>0$. Conversely, if every σ has some $t_x(\sigma)\ge k+1$, then that factor $(k)_{t_x(\sigma)}=0$ (the falling factorial vanishes once its length exceeds k), so every summand — and hence $\Omega_k(H)$ — is 0. □ □

Corollary 7.2 (Instances)**.** *Machine-verified instances (Section 9):*

$$\Omega_k(K_3)=8k^3-6k^2,\qquad \Omega_k(C_4)=2k^2(8k^2-8k+1),\qquad \Omega_k(K_{1,m})=\sum_j\binom{m}{j}(k)_j k^{m-j},$$

and, as a boundary example, $\Omega_1(K_4-e)=0$ (no orientation of the 5-edge butterfly keeps every vertex's owned degree ≤ 1), consistent with $\Omega_1(K_3)=2>0$. Dividing by $|\mathrm{Aut}(H)|$ gives the unlabelled-limit; e.g. $\lim\mathbb{E}T_{K_3}^{unlabelled}=(8k^3-6k^2)/6=(4k^3-3k^2)/3$ as in Theorem 4.2.

Remark. (a) The orientation invariant makes the star constants transparent ($\Omega_k(K_{1,m}) = c_m(k)$, Theorem 6.1). It also illustrates the budget-jam threshold: $\Omega_k(H)$ contrasts sharply with $G(n,p)$, where the leading behaviour for fixed H at matched density is $q^e n^{v-e}$, i.e. an $(2k/n)^e$-type scale: the k-out analogue $\Omega_k(H)$ *cannot* grow beyond what the per-vertex budget k can support in owned-degree terms, which is a strictly non-ER criterion — e.g. at $k=1$, K_4 is impossible ($\Omega_1(K_4)=0$), while K_3 has $\Omega_1(K_3)=2$ copies-of-kind asymptotics. (b) When $k \geq \Delta(H)$ (max degree), the identity-assignment (σ = all edges owned by their "first" endpoint under some orientation of H where each vertex owns $\leq \Delta(H) \leq k$) works, so $\Omega_k(H) > 0$ always for fixed H and $k \geq \Delta(H)$.

8 Exact second moment of the triangle count

Let T again be the (unlabelled) triangle count.

Theorem 8.1 (Exact first two moments)**.** *For all (n,k), exactly*

$$\mathbb{E}T^2 \;=\; \binom{n}{3}P_3 \;+\; 12\binom{n}{4}P_{K_4-e} \;+\; n\binom{n-1}{2}\binom{n-3}{2}P_{bowtie} \;+\; \binom{n}{3}\binom{n-3}{3}P_{2\triangle},$$

where $P_3 = P(\triangle)$ (Thm. 4.1) and P_{K_4-e}, P_{bowtie}, $P_{2\triangle}$ are the Theorem 3.1 probabilities of the three union graphs*: two triangles sharing an edge (K_4 minus an edge), sharing a vertex (the bowtie), and disjoint. In particular* $\mathrm{Var}(T) = \mathbb{E}T^2 - (\mathbb{E}T)^2$ *is an exact rational function of (n,k), and as $n \to \infty$ with fixed k:*

$$\mathrm{Var}(T) = v_0 + \frac{v_1}{n} + O(n^{-2}), \qquad v_0 = \frac{4k^3-3k^2}{3}, \quad v_1 = -\frac{k^2(16k^2-21k+7)}{2},$$

so that $\mathrm{Var}(T) \to \mathbb{E}T$ *(Var = mean asymptotically) with negative $1/n$-correction $v_1 < 0$ (underdispersion at every finite n staged, Table 5).*

Proof. T counts unordered triples; writing $T = \sum_s \mathbf{1}_s$ over the $\binom{n}{3}$ triples, we get $\mathbb{E}T^2 = \sum_{s,t}\Pr(E(s)\cup E(t))$, the (ordered) pair classes handled by the *overlap-pattern decomposition* of the two triples: $s = t$ ($\binom{n}{3}$ pairs, union = one triangle); $|s\cap t| = 2$ ($12\binom{n}{4}$ ordered pairs; union = $K_4 - e$); $|s\cap t| = 1$ ($n\binom{n-1}{2}\binom{n-3}{2}$ pairs; union = bowtie); $|s\cap t| = 0$ ($\binom{n}{3}\binom{n-3}{3}$ pairs; union = two disjoint triangles). Each union's edge-present probability is a Theorem 3.1 value at the respective template sizes (3,5,6,6 edges on 3,4,5,6 vertices). The series coefficients were derived and checked symbolically (all identities machine-verified; the Var-series was additionally checked against exact Fraction arithmetic at $n = 400$). □ □

Conjecture 8.2 (Poisson-type limit)**.** *For every fixed $k \geq 1$, as $n \to \infty$, the triangle count T converges in distribution to Poisson$\left(\frac{4k^3-3k^2}{3}\right)$.*

We do not prove Conjecture 8.2; we establish only the two exact moments (Theorem 8.1), whose agreement with the Poisson moments (mean = variance asymptotically) is suggestive, and we checked by Monte Carlo that the exact mean and exact variance agree with the sampled moments in Table 4 to within $|z| \leq 1.43$ in all seven staged (n,k) cells (dispersion Var$/\mathbb{E}$ at $n = 200$: 0.985 at $k=1$, 0.966 at $k=2$, 0.947 at $k=3$). A proof presumably requires a third-moment method or the $G(n,p)$-style Janson machinery under the k-out correlated conditioning, and is left open.

9 Verification and reproducibility

All numeric claims in this paper are reproduced by the public repository (see below for the placement), which contains the source code, the generated data, and the verification documents. The repository's verification stack:

Table 4: Monte Carlo vs. exact moments (`results/monte_carlo.csv`; the sampling corrects for the symmetrization via an OR-merge before the matrix product, cross-checked against a naive Fisher-consistency sampler).

n	k	reps	MC mean	exact $\mathbb{E}$	z	MC var / exact var	z
50	2	20000	6.46805	6.45824	+0.58	5.679 / 5.687	−0.13
50	3	20000	25.32875	25.34068	−0.36	21.744 / 21.437	+1.41
100	2	20000	6.57570	6.56458	+0.63	6.154 / 6.133	+0.34
100	3	20000	26.21760	26.18013	+1.09	23.662 / 23.669	−0.03
200	3	20000	26.64355	26.59253	+1.43	25.333 / 25.182	+0.60
200	4	20000	67.80790	67.76868	+0.70	63.251 / 63.114	+0.22
500	3	4000	26.80175	26.83760	−0.44	26.404 / 26.234	+0.29

Table 5: Dispersion of the triangle count (`results/variance_table.csv`; k = 1,2 shown; $k = 3$ staged in the file).

k	n	Var /$\mathbb{E}$	(Var $-v_0$)n	v_1
1	200	0.98515	−0.9916	−1.0000
2	200	0.96558	−55.6546	−58.0000
1	1000	0.99701	−0.9983	−1.0000
2	1000	0.99286	−57.5259	−58.0000

1. **Symbolic**: `src/verify_symbolic.py` — 36 machine-verified identities over exact rational algebra (sympy), including Theorem 3.1 vs. closed forms for triangles/C_4/stars ($m \le 6$), the identity (3), $B = AC$, the $k = 1$ specials of Corollary 4.3, the series coefficients of Theorems 4.2 and 8.1 (with e_3-based truncation error bounds), the edge identity, the matched-density ratio constants of Proposition 5.1, and the orientation-invariant instances of Corollary 7.2.

2. **Exhaustive enumeration**: for small n, *all* $\prod_v \binom{n-1}{k}$ choice profiles are enumerated and the copy counts matched exactly (exact-Fraction or float; 25 staged cases; largest difference $2.2 \cdot 10^{-15}$; the $(n, k) = (5, 2)$ second-moment case yields $\mathrm{Var}(T) = \frac{15985}{5832}$ exactly on both sides).

3. **Monte Carlo**: at larger n, Fisher-consistent sampling (symmetrized by an explicit OR-merge) with the staged cells of Table 4: worst $|z| = 1.43$ on means, 1.41 on variances, over 4,000–20,000 replicates per cell.

4. **Unit tests**: `pytest` (`tests/test_kout.py`), 43 tests, all passing (as of the release date recorded in the repository).

Reproducibility statement. The repository contains: `src/` (formulae, closed forms, exact-Fraction paths, exhaustive enumeration, Monte Carlo); `results/*.csv` (all tables behind Figures/Tables in this paper); `research/` (the research process documents: candidate scoring, literature review, evidence ledger, novelty assessment, hostile review, limitations, journal selection); and `paper/` (this manuscript and its bibliography). Running `python src/experiments.py` regenerates every CSV from scratch (Monte Carlo seeded); running `python` with the script `src/verify_symbolic.py` re-runs the 36 symbolic checks; `pytest` re-runs the 43 tests. Software: Python 3.14 with numpy, sympy (exact arithmetic), pytest. No proprietary data are used; the literature checks cite publicly retrievable sources with their access dates.

10 Related literature, in more detail

Global k-out structure. Connectivity and Hamiltonicity of $G(n, k\text{-out})$ and close relatives: [2] (connectivity of random m-orientable graphs and digraphs, Combinatorica 2(4), 1982); the textbook chapter [1, Ch. 22] (connectivity, perfect matchings, Hamilton cycles, nearest-neighbour graphs; no subgraph-count content — we verified the section structure against the authors' full text, retrieved October 2026); finite-n connectivity bounds [10]; robustness and giant component [9]; k-out subgraphs of large hosts [3].

Triangle/clustering in correlated models. For random key graphs (Eschenauer–Gligor key pools), [7] proved a zero-one law for triangle presence and [8] computed the clustering coefficient and contrasted it with matched ER — the closest methodological precedent to our Section 5, but on a different model (uniform key pools produce a distinct dependence structure and a zero-one law, not our $1 - \frac{3}{4k}$ density-ratio effect). Clustered random graph constructions with explicit triangle stubs: [11]; epidemic percolation on such constructions: Sood's thesis [12] (triangles as inserted building blocks, "doubly-Poisson" pairwise joint distributions; the k-out model enters only via connectivity remarks — checked against the full text, retrieved October 2026). Clustering *index* estimates are given in [6] for Erdős–Rényi with simulation for regular, Barabási–Albert and Watts–Strogatz models; the k-out model does not appear.

$G(n, p)$ **baselines.** The expected numbers $\binom{n}{v}p^e$, the Poisson/normal dichotomy of subgraph counts [5], and the classical zero-one/threshold framework of [4] are the backdrop for our matched-density comparison; none of them applies verbatim because of Lemma 2.1.

What is not known (to us). Beyond the Poisson-type Conjecture 8.2: (a) exact higher moments or the full distribution of T; (b) second moments for general H; (c) regimes with $k = k(n) \to \infty$ (dense k-out), where new phenomena must appear since $\mathbb{E}T \sim \frac{(4/3)k^3}{\cdot}$ would need to grow and cross the $G(n, p)$ law; (d) clustering coefficients of the k-out model at fixed k (which are not the ratio $\rho_\triangle$ and involve degree conditioning).

11 Limitations

(1) Our distributional results are first/second moment only; Conjecture 8.2 is not proven. (2) All asymptotic statements assume fixed k and fixed H; k growing with n is open. (3) The general-m star constants are explicit polynomials up to $m = 6$ (tabulated); the general formula $\sum_j \binom{m}{j}(k)_j k^{m-j}$ is proven via Theorem 7.1 for all m. (4) The orientation-invariant proof is complete, but its instance checks in the repository cover structured families (K_3, C_4, stars, $K_4 - e$); a general-implementation cross-check for arbitrary H was not built in-session. (5) The novelty statement (Section 9) is a negative claim from a targeted search (documents in the repository), not a proof of no-overlap: residual risks are recorded in the evidence ledger.

Data and code availability

All code, data (CSV), and research documentation are in the public repository accompanying this work (see the reproducibility statement in Section 9).

Competing interests

None. This work received no funding.

A Star constants $c_m(k)$ for $m \le 6$

$$
\begin{aligned}
c_2(k) &= k\,(4k-1),\\
c_3(k) &= 2k\,(4k^2-3k+1),\\
c_4(k) &= k\,(16k^3-24k^2+19k-6),\\
c_5(k) &= 2k\,(16k^4-40k^3+55k^2-40k+12),\\
c_6(k) &= k\,(64k^5-240k^4+500k^3-615k^2+418k-120).
\end{aligned}
$$

Each admits the structural form $c_m(k) = \sum_{j=0}^{m} \binom{m}{j}(k)_j\, k^{m-j}$ (Theorem 7.1) and the tabulated closed forms agree with it exactly (machine-checked for $m = 2, \ldots, 6$). Values at $k = 2, 3$ (used in Table 3): $c_2 : 14, 33$; $c_3 : 44, 168$; $c_4 : 128, 801$; $c_5 : 352, 3618$; $c_6 : 928, 15633$.

B Verification protocol details

The full verification documents (candidate scoring, literature review, evidence ledger, novelty assessment, two hostile-review passes, limitations, journal selection) accompany the repository under `research/`. The evidence ledger records, per claim: the theorem number, the check that confirms it (symbolic / enumeration / Monte Carlo / exact spot values), and any residual risks about the literature survey. Two independent hostile passes over the manuscript were recorded (pass 1: correctness-oriented attacks; pass 2: significance- and reproducibility-oriented attacks), including the discovery and repair of a Monte Carlo over-counting bug (documented; all Monte Carlo rows were regenerated after the fix).

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